

IZDANJE	IZMJENE I DOPUNE	IZMJENIO	PREGLED. I ODOBRIO



ULJANIK
Brodogradilište, d.d.
Pula, Croatia

PROJEKTIRAO

Brkić Irena

PREGLEDAO

Pašuld Jasna

ODOBRIO

Pavletić Renis

STROJARSKA OPREMA - TEHNIČKI PODACI

	N A C R T				GRADNJA BR. - YARD No:				List Sheet	ODOBRE Optegra V	
	1-106-522				503						
MF:	IZDANJE - ISSUE	A		B		C	D	E	F	G	H
	DATUM - DATE : 09.04.14										



**ULJANIK****STROJARSKA OPREMA - TEHNIČKI
PODACI**Nacrt
1-106-522OJ
8C42Gradnja **503**

Str 1/47, 16.05.2014

Sistem **441143**
BRUSILICA

Oznaka	Naziv opreme	JM	Kol.
441143-4-001	BRUSILICA S UGRAĐENIM PRIBOROM 2X 150X20	KOM	1

Dobavljač **ISTRATEHNA D.O.O.**
 Katuri 24
 52220 LABIN
 HRV Republika Hrvatska
 tel. 855-243 fax.

Broj i datum narudžbe: **292511** / 08.05.2013

Proizvođač **ISTRATEHNA D.O.O.**
 Katuri 24
 52220 LABIN
 HRV Republika Hrvatska
 tel. 855-243 telefax.

Cert	Tip certifikata	Clas.	Broj mape
	Broj certifikata	Broj uredaja	
C10	Asbestos Free Certificate DECLARATION OF CONFORMITY		
C4	Works Certificate "EC"DECLARATION OF CONFORMITY		

Naziv knjiga uputa	Map no.
GRINDING MACHINE	1

Tehnički podaci

Double side grinder of column type for fixing onto floor provided with:

- shaft with two ends to be used for grinding plate fixing
- grinding wheels of 125 mm diameter
- electric equipment with driving single-speed motor for 220 V, 50 Hz voltage with belonging starting and protection elements
- the el. motor to be single speed, squirrel-cage type for direct starting, IP-54 enclosure, marine tropic design, class insulation: "F", for continuous loaded drive at an ambient temp. of 45 dgr. C and 90 % humidity, equipped with a connecting box for cable, of IP-44 enclosure type
- with protection against overloading, under voltage and short circuit

		STROJARSKA OPREMA - TEHNIČKI PODACI		Nacrt 1-106-522	OJ 8C42
				Gradnja 503	Str 2/47, 16.05.2014
Sistem 581171 CENTRIFUGALNE PUMPE - SANITARNA VODA					
Oznaka 581171-2-001	Naziv opreme PUMPA SANITARNE VODE 10 M3/H; 6 BAR			JM KOM	Kol. 1
Dobavljač TTK - TVORNICA TURBINA D.O.O. Vrazova 53, P.P. 168 47000 KARLOVAC HRV Republika Hrvatska tel. 047/648-571 fax. Broj i datum narudžbe: 293759 / 03.07.2013					
Proizvođač TTK - TVORNICA TURBINA D.O.O. Vrazova 53, P.P. 168 47000 KARLOVAC HRV Republika Hrvatska tel. 047/648-571 telefax.					
Cert C4	Tip certifikata Broj certifikata Works Certificate 1153-3/296988		Clas. Broj uredaja	Broj mape 1	
Naziv knjiga uputa PUMPE STROJARNICE-FINALNA DOKUMEN					Map no. 1
Tehnički podaci CENTRIFUGAL PUMP WITH EL. MOTOR ----- MARK:..... 581171-2-001 PIECES:..... 1 (ONE) CAPACITY:..... 10 m3/h TOTAL MANOMETRIC HEAD:.. 6 Bar SERVICE:..... potable water, drencher DESCRIPTION: vertical, with intermediate distance shaft SPEED..... * rpm SELPRIMING..... YES LIQUID:..... fresh water SPECIFIC GRAVITY: 998 kg/m3 KIN. VISCOSITY:..... 0,8 mm2/s WORKING TEMPERATURE:cca. 10-40 deg.C NPSH available 4,5 mlc MATERIAL OF CASING..... bronze MATERIAL OF impeller..... bronze MATERIAL OF SHAFT:..... stainless steel BEARINGS:..... ball bearings SUCTION FLANGE acc.to DIN 2533; DN * / NP 16 DELIVERY FLANGE acc.to DIN 2533; DN * / NP 16					

**ULJANIK****STROJARSKA OPREMA - TEHNIČKI
PODACI**Nacrt
1-106-522OJ
8C42Gradnja **503**

Str 3/47, 16.05.2014

Oznaka	Naziv opreme	JM	Kol.
581171-2-001	PUMPA SANITARNE VODE 10 M3/H; 6 BAR	KOM	1

WEIGHT OF PUMP WITH EL. MOTOR:.....*.. kg
CERTIFICATE FOR PUMPCRS
CERTIFICATE FOR EL. MOTOR:..... manufacturer's
INSULATION CLASS :..... IP 55
PROTECTION : F
ANTICONDENZATING HEATER : NO

**ULJANIK****STROJARSKA OPREMA - TEHNIČKI
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1-106-522OJ
8C42Gradnja **503**

Str 4/47, 16.05.2014

Sistem **631661**
SKLOP PROPULZIONE JEDINICE

Oznaka	Naziv opreme	JM	Kol.
631661-1-010	OKRETNi PROPULZOR, LIJEVOKRETNi	KOM	2

Dobavljač **SCHOTTEL GMBH**
Mainzer Strasse 99
56322 SPAY/RHEIN
DEU Germany
tel. 00492628 61 0 fax.

Broj i datum narudžbe: **292529** / 30.04.2013

Proizvođač **SCHOTTEL GMBH**
Mainzer Strasse 99
56322 SPAY/RHEIN
DEU Germany
tel. 00492628 61 0 telefax.

Cert	Tip certifikata Broj certifikata	Clas. Broj uredaja	Broj mape
C1	Classification Certificate 39349 D	HRB 2819,2829,2821,2822	1

Naziv knjiga uputa	Map no.
RUDDER PROPELLER Vol.1	2
RUDDER PROPELLER Vol.2	3

Tehnički podaci

Azimuth thruster with FP propeller
Nominal power 500 kW - Continuos duty 24/24 hrs - 600-1800 rpm
Propeller design - 80% MCR / 1800 rpm - ship speed 12,5 kn / draught 2,2m
Propeller diameter - abt 1400mm
Turning direction of Input shaft seen on input shaft - CLOCKWISE
Turning direction of propeller - COUNTERCLOCKWISE

Steering
rotation speed - aprox 12 sec/180 deg
hydraulic pump power supply - main engine PTO

**ULJANIK****STROJARSKA OPREMA - TEHNIČKI
PODACI**Nacrt
1-106-522OJ
8C42Gradnja **503**

Str 5/47, 16.05.2014

Oznaka	Naziv opreme	JM	Kol.
631661-1-020	OKRETNi PROPULZOR, DESNOKRETNi	KOM	2

Dobavljač **SCHOTTEL GMBH**
 Mainzer Strasse 99
 56322 SPAY/RHEIN
 DEU Germany
 tel. 00492628 61 0 fax.
 Broj i datum narudžbe: **292529** / 30.04.2013

Proizvođač **SCHOTTEL GMBH**
 Mainzer Strasse 99
 56322 SPAY/RHEIN
 DEU Germany
 tel. 00492628 61 0 telefax.

Cert	Tip certifikata	Clas.	Broj mape
	Broj certifikata	Broj uredaja	
C1	Classification Certificate	HRB	
	39349D	2819,2829,2821,2822	1

Naziv knjiga uputa	Map no.
RUDDER PROPELLER Vol.1	2
RUDDER PROPELLER Vol.2	3

Tehnički podaci

Azimuth thruster with FP propeller
 Nominal power 500 kW - Continuos duty 24/24 hrs - 600-1800 rpm
 Propeller design - 80% MCR / 1800 rpm - ship speed 12,5 kn / draught 2,2m
 Propeller diameter - abt 1400mm
 Turning direction of Input shaft seen on input shaft - CLOCKWISE
 Turning direction of propeller - CLOCKWISE

Steering
 rotation speed - aprox 12 sec/180 deg
 hydraulic pump power supply - main engine PTO

Naziv datoteke (PDM/UCM): 503-8c42-1106522-o-1.pdf

		STROJARSKA OPREMA - TEHNIČKI PODACI		Nacrt 1-106-522		OJ 8C42	
				Gradnja 503		Str 6/47, 16.05.2014	
Sistem 701173 PUMPE VIJČANE - PREBACIVANJE GORIVA							
Oznaka		Naziv opreme				JM	
701173-2-001		PUMPA TRANSFERA GORIVA - PRAMČANA 4 M3/H				KOM	
Kol.		1					
Dobavljač MPD TVORNICA PUMPI D.D. Petra Preradovića 73 43500 DARUVAR HRV Republika Hrvatska tel. 043/331-444 fax. Broj i datum narudžbe: 293459 / 02.07.2013							
Proizvođač MPD TVORNICA PUMPI D.D. Petra Preradovića 73 43500 DARUVAR HRV Republika Hrvatska tel. 043/331-444 telefax.							
Cert		Tip certifikata		Clas.		Broj mape	
		Broj certifikata		Broj uredaja			
C1		Classification Certificate 20-053652/2011520		HRB 138623 ; 138624		1	
Naziv knjiga uputa						Map no.	
PUMPA TRANSFER GORIVA						4	
Tehnički podaci							
SCREW PUMP WITH ELECT. MOTOR							

1. MARK:..... 701173-2-001 2. PIECE:..... one (1) 3. SERVICE:..... FUEL 4. DESCRIPTION: horizontal with mechanical seal 5. BUILT IN BY-PASS RELIEF VALVE:..... YES 6. LIQUID:..... MARINE DIESEL OIL 7. SPECIFIC GRAVITY: 860 kg/m3 8. KIN. VISCOSITY:..... 25mm2/s 9. WORKING TEMPERATURE: 20-50 deg.C 10. CAPACITY:..... 4m3/h 11. TOTAL MANOMETRIC HEAD:..... 3,5 bar (pressure 3bar; suction -05 bar min.) 12. NPSH available.....abt.3,7 M.L.C. If is liquid level under pump. 13. SUCTION HEAD (PRESSURE):.min. -0,5 bar If is liquid level above pump. Difference between geodetic suction head and friction losses in suction pipe. 14. NPSH:.....*.. MLC 15. TYPE:.....*.. 16. NO. OF DRIVEN SCREW:.....*.. 17. SPEED:..... 1500 RPM or less							



Oznaka	Naziv opreme	JM	Kol.
701173-2-001	PUMPA TRANSFERA GORIVA - PRAMČANA 4 M3/H	KOM	1

18. SHAFT NECESSARY POWER:.....*.. KW
19. INSTALLED POWER:.....*.. KW
20. SUCTION FLANGE acc.to DIN 2533;DN * / NP 16
21. DELIVERY FLANGE acc.to DIN 2533;DN * / NP 16
22. MATERIAL OF CASING:.....cast iron
23. MATERIAL OF SCREWS:.....stainless steel
24. MATERIAL OF SHAFT:.....stainless steel
25. GAUGE CONNECTION INCLUDED:.....*..
26. WEIGHT OF PUMP WITH EL. MOTOR:.....*.. kg
27. CERTIFICATE FOR PUMP CRS
28. CERTIFICATE FOR ELECTRIC MOTOR MANUFACTURER

**ULJANIK****STROJARSKA OPREMA - TEHNIČKI
PODACI**Nacrt
1-106-522Gradnja **503**OJ
8C42

Str 8/47, 16.05.2014

Oznaka	Naziv opreme	JM	Kol.
701173-2-002	PUMPA TRANSFERA GORIVA - KRMENA 4 M3/H;	KOM	1

Dobavljač **MPD TVORNICA PUMPI D.D.**

Petra Preradovića 73

43500 DARUVAR

HRV Republika Hrvatska

tel. 043/331-444 fax.

Broj i datum narudžbe: **293459** / 02.07.2013Proizvođač **MPD TVORNICA PUMPI D.D.**

Petra Preradovića 73

43500 DARUVAR

HRV Republika Hrvatska

tel. 043/331-444 telefax.

Cert	Tip certifikata Broj certifikata	Clas. Broj uredaja	Broj mape
C1	Classification Certificate 20-053652/2011520	HRB 138623 ; 138624	1

Naziv knjiga uputa	Map no.
PUMPA TRANSFER GORIVA	4

Tehnički podaci

SCREW PUMP WITH ELECT. MOTOR

1. MARK:..... 701173-2-002
2. PIECE:..... one (1)
3. SERVICE:..... FUEL
4. DESCRIPTION: horizontal with mechanical seal
5. BUILT IN BY-PASS RELIEF VALVE:..... YES
6. LIQUID:..... MARINE DIESEL OIL
7. SPECIFIC GRAVITY: 860 kg/m³
8. KIN. VISCOSITY:..... 25 mm²/s
9. WORKING TEMPERATURE: 20-50 deg.C
10. CAPACITY:..... 4m³/h
11. TOTAL MANOMETRIC HEAD:..... 3,5 bar (pressure 3bar; suction -05 bar min.)
12. NPSH available.....abt.3,7 M.L.C.
If is liquid level under pump.
13. SUCTION HEAD (PRESSURE):.min. -0,5 bar
If is liquid level above pump.
Difference between geodetic suction head and friction losses in suction pipe.
14. NPSH:.....*.. MLC
15. TYPE:.....*..
16. NO. OF DRIVEN SCREW:.....*..
17. SPEED:..... 1500 RPM or less
18. SHAFT NECESSARY POWER:.....*.. KW
19. INSTALLED POWER:.....*.. KW

**ULJANIK****STROJARSKA OPREMA - TEHNIČKI
PODACI**Nacrt
1-106-522OJ
8C42Gradnja **503**

Str 9/47, 16.05.2014

Oznaka	Naziv opreme	JM	Kol.
701173-2-002	PUMPA TRANSFERA GORIVA - KRMENA 4 M3/H;	KOM	1

20. SUCTION FLANGE acc.to DIN 2533;DN * / NP 16
21. DELIVERY FLANGE acc.to DIN 2533;DN * / NP 16
22. MATERIAL OF CASING:.....cast iron
23. MATERIAL OF SCREWS:.....stainless steel
24. MATERIAL OF SHAFT:.....stainless steel
25. GAUGE CONNECTION INCLUDED:.....*..
26. WEIGHT OF PUMP WITH EL. MOTOR:.....*.. kg
27. CERTIFICATE FOR PUMP CRS
28. CERTIFICATE FOR ELECTRIC MOTOR MANUFACTURER

		STROJARSKA OPREMA - TEHNIČKI PODACI		Nacrt 1-106-522	OJ 8C42
				Gradnja 503	Str 10/47, 16.05.2014
Sistem 711173 PUMPE VIJČANE - PREBACIVANJE MAZIVA					
Oznaka 711173-2-001	Naziv opreme TRANSFER PUMPA PRLJAVOG ULJA-PRAMČANA 4				JM KOM Kol. 1
Dobavljač MPD TVORNICA PUMPI D.D. Petra Preradovića 73 43500 DARUVAR HRV Republika Hrvatska tel. 043/331-444 fax. Broj i datum narudžbe: 293459 / 02.07.2013					
Proizvođač MPD TVORNICA PUMPI D.D. Petra Preradovića 73 43500 DARUVAR HRV Republika Hrvatska tel. 043/331-444 telefax.					
Cert C1	Tip certifikata Broj certifikata Classification Certificate 20-053651/2011520		Clas. Broj uredaja HRB 138627 ; 138628	Broj mape 1	
Naziv knjiga uputa TRANSFER PUMPA PRLJAVOG ULJA					Map no. 4
Tehnički podaci SCREW PUMP WITH ELECT. MOTOR ----- 1. MARK:.....711173-2-001 2. PIECE:..... one (1) 3. SERVICE:..... LUB.OIL 4. DESCRIPTION: vertical/ with mechanical seal 5. BUILT IN BY-PASS RELIEF VALVE:..... YES 6. LIQUID:..... LUB.OIL 7. SPECIFIC GRAVITY:at 15 deg.C 900 kg/m3 8. KIN. VISCOSITY:.....at 50 deg.C 75 mm2/s 9. WORKING TEMPERATURE: 20-50 deg.C 10. CAPACITY:..... 4 m3/h 11. TOTAL MANOMETRIC HEAD:..... 3,5 bar (pressure 3bar ; suction -0,5bar min.) 12. NPSH available...abt. 3,7 M.L.C. If is liquid level under pump. 13. SUCTION HEAD (PRESSURE):.....min. -0,5 bar If is liquid level above pump. Difference between geodetic suction head and friction losses in suction pipe. 14. NPSH:.....*.. MLC 15. TYPE:.....*.. 16. NO. OF DRIVEN SCREW:.....*.. 17. SPEED:..... 1500 RPM or less 18. SHAFT NECESSARY POWER:.....*.. KW					

**ULJANIK****STROJARSKA OPREMA - TEHNIČKI
PODACI**Nacrt
1-106-522OJ
8C42Gradnja **503**

Str 11/47, 16.05.2014

Oznaka	Naziv opreme	JM	Kol.
711173-2-001	TRANSFER PUMPA PRLJAVOG ULJA-PRAMČANA 4	KOM	1

18. SHAFT NECESSARY POWER:.....*.. KW
19. INSTALLED POWER:.....*.. KW
20. SUCTION FLANGE acc.to DIN 2533;DN */ NP 16
21. DELIVERY FLANGE acc.to DIN 2533;DN */ NP 16
22. MATERIAL OF CASING:..... cast iron
23. MATERIAL OF SCREWS:..... stainless steel
24. MATERIAL OF SHAFT:..... stainless steel
25. GAUGE CONNECTION INCLUDED:.....*..
26. WEIGHT OF PUMP WITH EL. MOTOR:.....*.. kg
27. CERTIFICATE FOR PUMP..... CRS
28. CERTIFICATE FOR ELECTRIC MOTOR..... MANUFACTURERS

**ULJANIK****STROJARSKA OPREMA - TEHNIČKI
PODACI**Nacrt
1-106-522OJ
8C42Gradnja **503**

Str 12/47, 16.05.2014

Oznaka	Naziv opreme	JM	Kol.
711173-2-002	TRANSFER PUMPA PRLJAVOG ULJA-KRMENA 4 M3	KOM	1

Dobavljač **MPD TVORNICA PUMPI D.D.**

Petra Preradovića 73

43500 DARUVAR

HRV Republika Hrvatska

tel. 043/331-444 fax.

Broj i datum narudžbe: **293459** / 02.07.2013Proizvođač **MPD TVORNICA PUMPI D.D.**

Petra Preradovića 73

43500 DARUVAR

HRV Republika Hrvatska

tel. 043/331-444 telefax.

Cert	Tip certifikata Broj certifikata	Clas. Broj uredaja	Broj mape
C1	Classification Certificate 20-053651/2011520	HRB 138627 ; 138628	1

Naziv knjiga uputa	Map no.
TRANSFER PUMPA PRLJAVOG ULJA	4

Tehnički podaci

SCREW PUMP WITH ELECT. MOTOR

-
- MARK:.....711173-2-002
 - PIECE:..... one (1)
 - SERVICE:..... LUB.OIL
 - DESCRIPTION: vertical/ with mechanical seal
 - BUILT IN BY-PASS RELIEF VALVE:..... YES
 - LIQUID:..... LUB.OIL
 - SPECIFIC GRAVITY:at 15 deg.C 900 kg/m3
 - KIN. VISCOSITY:.....at 50 deg.C 75 mm2/s
 - WORKING TEMPERATURE: 20-50 deg.C
 - CAPACITY:..... 4 m3/h
 - TOTAL MANOMETRIC HEAD:..... 3,5 bar (pressure 3bar ; suction -0,5bar min.)
 - NPSH available...abt. 3,7 M.L.C.
If is liquid level under pump.
 - SUCTION HEAD (PRESSURE):.....min. -0,5 bar
If is liquid level above pump.
Difference between geodetic suction head and friction losses in suction pipe.
 - NPSH:.....*.. MLC
 - TYPE:.....*..
 - NO. OF DRIVEN SCREW:.....*..
 - SPEED:..... 1500 RPM or less
 - SHAFT NECESSARY POWER:.....*.. KW
 - SHAFT NECESSARY POWER:.....*.. KW
 - INSTALLED POWER:.....*.. KW

**ULJANIK****STROJARSKA OPREMA - TEHNIČKI
PODACI**Nacrt
1-106-522OJ
8C42Gradnja **503**

Str 13/47, 16.05.2014

Oznaka	Naziv opreme	JM	Kol.
711173-2-002	TRANSFER PUMPA PRLJAVOG ULJA-KRMENA 4 M3	KOM	1

20. SUCTION FLANGE acc.to DIN 2533;DN * / NP 16
21. DELIVERY FLANGE acc.to DIN 2533;DN * / NP 16
22. MATERIAL OF CASING:..... cast iron
23. MATERIAL OF SCREWS:..... stainless steel
24. MATERIAL OF SHAFT:..... stainless steel
25. GAUGE CONNECTION INCLUDED:.....*..
26. WEIGHT OF PUMP WITH EL. MOTOR:.....*.. kg
27. CERTIFICATE FOR PUMP..... CRS
28. CERTIFICATE FOR ELECTRIC MOTOR..... MANUFACTURERS

**ULJANIK****STROJARSKA OPREMA - TEHNIČKI
PODACI**Nacrt
1-106-522Gradnja **503**OJ
8C42

Str 14/47, 16.05.2014

Sistem **721263**
UREĐAJ PROTIV OBRAŠTANJA - AKTIVNI HIPOKLORIT

Oznaka	Naziv opreme	JM	Kol.
721263-0-001	ELEKTRO PANEL R200	CPL	1

Dobavljač **AZIENDA CHIMICA GENOVESE S.R.L**
Via F. Vezzani 18
16159 GENOVA
ITA Italy
tel. (010)-442694 fax.
Broj i datum narudžbe: **293008** / 29.05.2013

Proizvođač **AZIENDA CHIMICA GENOVESE S.R.L**
Via F. Vezzani 18
16159 GENOVA
ITA Italy
tel. (010)-442694 telefax.

Cert	Tip certifikata Broj certifikata	Clas. Broj uredaja	Broj mape
C1	Classification Certificate 2013/MI/01/680-1	HRB	1
C4	Works Certificate WORKSHOP CERTIFICATE		

Naziv knjiga uputa	Map no.
ECOLCELL AF R200+ECOLCELL R100	4

Tehnički podaci
TYPE R100/R200 WITH PUMP STARTER

**ULJANIK****STROJARSKA OPREMA - TEHNIČKI
PODACI**Nacrt
1-106-522OJ
8C42Gradnja **503**

Str 15/47, 16.05.2014

Sistem **733165**
KOMPRES.RADNOG ZRAKA,ROTACIONI

Oznaka	Naziv opreme	JM	Kol.
733165-2-001	KOMPRESOR RADNOG ZRAKA 50M3/H FA.D. / 7	KOM	2

Dobavljač **SPERRE INDUSTRI A/S**
 Sperre
 6057 ELLINGSOY
 NOR Norveška
 tel. 0047-70-16-11-00 fax.
 Broj i datum narudžbe: **292545** / 08.07.2013

Proizvođač **SPERRE INDUSTRI A/S**
 Sperre
 6057 ELLINGSOY
 NOR Norveška
 tel. 0047-70-16-11-00 telefax.

Cert	Tip certifikata Broj certifikata	Clas. Broj uredaja	Broj mape
C1	Classification Certificate ALE 13 3318 ALE 13 3317	HRB 95281 95280	
C4	Works Certificate TEST RECORD		

Naziv knjiga uputa	Map no.
AIR COMPRESSOR	4

Tehnički podaci

Piston type, driven by three-phase electromotor.
 Suction volume: 50 m3/h (f.a.d.)
 Working pressure: 7 bar
 Cooling: by air

The compressors should be fitted with standard accessories:
 suction filter, safety valves,, attached pump for lubrication of all bearings, intercooler, aftercooler, oil and water separator, solenoid water stop valve, temp. indication and alarm (set point as per rules or defined later)
 The compressors should be provided with automatic condensate drain, auto safety device
 Alarm indication for low lub oil pressure and high discharge air temperature.
 All connection pipes of compressors have to be flexibles (Class certificate)

The compressors should be prepared for remote control and stopping
 at a defined pressure as well as for continuous running with
 periodical automatic unloadings depending upon pressure in air vessel.

**ULJANIK****STROJARSKA OPREMA - TEHNIČKI
PODACI**Nacrt
1-106-522OJ
8C42Gradnja **503**

Str 16/47, 16.05.2014

Oznaka	Naziv opreme	JM	Kol.
733165-2-001	KOMPRESOR RADNOG ZRAKA 50M3/H FA.D. / 7	KOM	2

Electro-motor and compressor to be fitted on common foundation, interconnected by means of an elastic coupling.
- Available supply: 3 x400 V,50 Hz as well as 220V, 50Hz ,
El. motor efficiency grade: IE3

The unit will be installed on elastic foundation

**ULJANIK****STROJARSKA OPREMA - TEHNIČKI
PODACI**Nacrt
1-106-522OJ
8C42Gradnja **503**

Str 17/47, 16.05.2014

Sistem **790876**
TERMOMETRI

Oznaka	Naziv opreme	JM	Kol.
790876-2-023	TERMOMETAR RAVNI 0-60°C,G=1/2,L1=63MM	KOM	2

Dobavljač **INENCO D.O.O.**
V. Ruždjaka 11
10000
HRV Republika Hrvatska
tel. 01/611-9613 fax.
Broj i datum narudžbe: **295692** / 23.09.2013Proizvođač **INENCO D.O.O.**
V. Ruždjaka 11
10000
HRV Republika Hrvatska
tel. 01/611-9613 telefax.

Cert	Tip certifikata Broj certifikata	Clas. Broj uredaja	Broj mape
C2	Type Approval Classification Certificate 13 676 - 99 HH		
C4	Works Certificate EC-Declaration of Conformity		

Naziv knjiga uputa	Map no.
TERMOMETERS	4

Tehnički podaci

**ULJANIK****STROJARSKA OPREMA - TEHNIČKI
PODACI**Nacrt
1-106-522OJ
8C42Gradnja **503**

Str 18/47, 16.05.2014

Oznaka	Naziv opreme	JM	Kol.
790876-2-130	TERMOMETAR CAPILARNI 0-60 C,G=1/2,L1=160	KOM	2

Dobavljač **INENCO D.O.O.**
V. Ruždjaka 11
10000
HRV Republika Hrvatska
tel. 01/611-9613 fax.
Broj i datum narudžbe: **295692** / 23.09.2013

Proizvođač **INENCO D.O.O.**
V. Ruždjaka 11
10000
HRV Republika Hrvatska
tel. 01/611-9613 telefax.

Cert	Tip certifikata Broj certifikata	Clas. Broj uredaja	Broj mape
C4	Works Certificate EC-Declaration of Conformity		

Naziv knjiga uputa	Map no.
THERMOMETERS	4

Tehnički podaci

- nominal size 100(diameter)
- material stainless steel
- capillary lenght 2 m
- surface mounting flange,stainless steel
- process connection:G 1/2 B
- stem diameter:6mm
- stem lenght:160mm
- scale range:0-60 C

**ULJANIK****STROJARSKA OPREMA - TEHNIČKI
PODACI**Nacrt
1-106-522OJ
8C42Gradnja **503**

Str 19/47, 16.05.2014

Sistem **790881**
NIVOSTATI

Oznaka	Naziv opreme	JM	Kol.
790881-2-120	NIVOSTAT	KOM	16

Dobavljač **INELTEH - INŽINJERING ZA ELEKT**
Ljubljanska cesta 7
51000 RIJEKA
HRV Republika Hrvatska
tel. 051/274-536 fax.
Broj i datum narudžbe: **295114** / 07.08.2013

Proizvođač **INELTEH - INŽINJERING ZA ELEKT**
Ljubljanska cesta 7
51000 RIJEKA
HRV Republika Hrvatska
tel. 051/274-536 telefax.

Cert	Tip certifikata Broj certifikata	Clas. Broj uredaja	Broj mape
C2	Type Approval Classification Certificate 03-001315/030990		
C4	Works Certificate Rijeka,20.02.2014		

Naziv knjiga uputa	Map no.
MAGNETIC LEVEL SWITCHES	4

Tehnički podaci

**ULJANIK****STROJARSKA OPREMA - TEHNIČKI
PODACI**Nacrt
1-106-522Gradnja **503**OJ
8C42

Str 20/47, 16.05.2014

Oznaka	Naziv opreme	JM	Kol.
790881-2-121	NIVOSTAT	KOM	2

Dobavljač **INELTEH - INŽINJERING ZA ELEKT**
Ljubljanska cesta 7
51000 RIJEKA
HRV Republika Hrvatska
tel. 051/274-536 fax.
Broj i datum narudžbe: **295114** / 07.08.2013

Proizvođač **INELTEH - INŽINJERING ZA ELEKT**
Ljubljanska cesta 7
51000 RIJEKA
HRV Republika Hrvatska
tel. 051/274-536 telefax.

Cert	Tip certifikata	Clas.	Broj mape
	Broj certifikata	Broj uredaja	
C2	Type Approval Classification Certificate	03-001315/030990	
C4	Works Certificate Rijeka,20.02.2014.		

Naziv knjiga uputa	Map no.
MAGNETIC LEVEL SWITCHES	4

Tehnički podaci

Naziv datoteke (PDM/UCM): 503-8c42-1106522-o-1.pdf

**ULJANIK****STROJARSKA OPREMA - TEHNIČKI
PODACI**Nacrt
1-106-522OJ
8C42Gradnja **503**

Str 21/47, 16.05.2014

Oznaka	Naziv opreme	JM	Kol.
790881-2-128	NIVOSTAT	KOM	10

Dobavljač **INELTEH - INŽINJERING ZA ELEKT**
Ljubljanska cesta 7
51000 RIJEKA
HRV Republika Hrvatska
tel. 051/274-536 fax.
Broj i datum narudžbe: **295114** / 07.08.2013

Proizvođač **INELTEH - INŽINJERING ZA ELEKT**
Ljubljanska cesta 7
51000 RIJEKA
HRV Republika Hrvatska
tel. 051/274-536 telefax.

Cert	Tip certifikata Broj certifikata	Clas. Broj uredaja	Broj mape
C2	Type Approval Classification Certificate 03-001315/030990		
C4	Works Certificate Rijeka,20.02.2014.		

Naziv knjiga uputa	Map no.
MAGNETIC LEVEL SWITCHES	4

Tehnički podaci

Naziv datoteke (PDM/UCM): 503-8c42-1106522-o-1.pdf

**ULJANIK****STROJARSKA OPREMA - TEHNIČKI
PODACI**Nacrt
1-106-522OJ
8C42Gradnja **503**

Str 22/47, 16.05.2014

Oznaka	Naziv opreme	JM	Kol.
790881-2-129	NIVOSTAT	KOM	4

Dobavljač **INELTEH - INŽINJERING ZA ELEKT**
Ljubljanska cesta 7
51000 RIJEKA
HRV Republika Hrvatska
tel. 051/274-536 fax.
Broj i datum narudžbe: **297283** / 25.10.2013

Proizvođač **INELTEH - INŽINJERING ZA ELEKT**
Ljubljanska cesta 7
51000 RIJEKA
HRV Republika Hrvatska
tel. 051/274-536 telefax.

Cert	Tip certifikata Broj certifikata	Clas. Broj uredaja	Broj mape
C2	Type Approval Classification Certificate 03-001315/030990		
C4	Works Certificate Rijeka,20.02.2014.		

Naziv knjiga uputa	Map no.
MAGNETIC LEVEL SWITCHES	4

Tehnički podaci

With junction box ITJB-1208,submerged,IP68,5m cable.

Naziv datoteke (PDM/UCM): 503-8c42-1106522-o-1.pdf

**ULJANIK****STROJARSKA OPREMA - TEHNIČKI
PODACI**Nacrt
1-106-522OJ
8C42Gradnja **503**

Str 23/47, 16.05.2014

Oznaka 790881-2-160	Naziv opreme NIVOSTAT DVOKONTAKTNI	JM KOM	Kol. 2
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Dobavljač **INELTEH - INŽINJERING ZA ELEKT**

Ljubljanska cesta 7

51000 RIJEKA

HRV Republika Hrvatska

tel. 051/274-536 fax.

Broj i datum narudžbe: **297283** / 25.10.2013Proizvođač **INELTEH - INŽINJERING ZA ELEKT**

Ljubljanska cesta 7

51000 RIJEKA

HRV Republika Hrvatska

tel. 051/274-536 telefax.

Cert	Tip certifikata Broj certifikata	Clas. Broj uredaja	Broj mape
C2	Type Approval Classification Certificate 03-001315/030990		
C4	Works Certificate Rijeka,20.02.2014.		

Naziv knjiga uputa MAGNETIC LEVEL SWITCHES	Map no. 4
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Tehnički podaci

With junction box ITJB-1208,submerged,IP68,5m cable.

Naziv datoteke (PDM/UCM): 503-8c42-1106522-o-1.pdf

**ULJANIK****STROJARSKA OPREMA - TEHNIČKI
PODACI**Nacrt
1-106-522OJ
8C42Gradnja **503**

Str 24/47, 16.05.2014

Oznaka	Naziv opreme	JM	Kol.
790881-2-161	NIVOSTAT DVOKONTAKTNI-DUZI	KOM	4

Dobavljač INELTEH - INŽINJERING ZA ELEKT

Ljubljanska cesta 7

51000 RIJEKA

HRV Republika Hrvatska

tel. 051/274-536 fax.

Broj i datum narudžbe: **297283** / 25.10.2013**Proizvođač INELTEH - INŽINJERING ZA ELEKT**

Ljubljanska cesta 7

51000 RIJEKA

HRV Republika Hrvatska

tel. 051/274-536 telefax.

Cert	Tip certifikata	Clas.	Broj mape
	Broj certifikata	Broj uredaja	
C2	Type Approval Classification Certificate 03-001315/030990		
C4	Works Certificate Rijeka,20.02.2014.		

Naziv knjiga uputa	Map no.
MAGNETIC LEVEL SWITCHES	4

Tehnički podaci

Naziv datoteke (PDM/UCM): 503-8c42-1106522-o-1.pdf

		STROJARSKA OPREMA - TEHNIČKI PODACI		Nacrt 1-106-522		OJ 8C42	
				Gradnja 503		Str 25/47, 16.05.2014	
Oznaka		Naziv opreme				JM	
790881-2-190		NIVOSTAT WITH PROTECTION TUBE				KOM	
Kol.		4					
Dobavljač INELTEH - INŽINJERING ZA ELEKT Ljubljanska cesta 7 51000 RIJEKA HRV Republika Hrvatska tel. 051/274-536 fax. Broj i datum narudžbe: 302410 / 06.05.2014							
Proizvođač INELTEH - INŽINJERING ZA ELEKT Ljubljanska cesta 7 51000 RIJEKA HRV Republika Hrvatska tel. 051/274-536 telefax.							
Cert	Tip certifikata			Clas.		Broj mape	
	Broj certifikata			Broj uredaja			
C2	Type Approval Classification Certificate 03-001315/030990					1	
C4	Works Certificate MANUFACT.CERT			051421249-051421252		1	
C6	EX Certificate IMQ 09 ATEX 013 X					1	
Naziv knjiga uputa						Map no.	
MAGNETIC LEVEL SWITCHES						4	
Tehnički podaci							

**ULJANIK****STROJARSKA OPREMA - TEHNIČKI
PODACI**Nacrt
1-106-522Gradnja **503**OJ
8C42

Str 26/47, 16.05.2014

Oznaka	Naziv opreme	JM	Kol.
790881-2-191	ZENER BARIJERA	KOM	2

Dobavljač **INELTEH - INŽINJERING ZA ELEKT**
Ljubljanska cesta 7
51000 RIJEKA
HRV Republika Hrvatska
tel. 051/274-536 fax.
Broj i datum narudžbe: **302410** / 06.05.2014

Proizvođač **INELTEH - INŽINJERING ZA ELEKT**
Ljubljanska cesta 7
51000 RIJEKA
HRV Republika Hrvatska
tel. 051/274-536 telefax.

Cert	Tip certifikata Broj certifikata	Clas. Broj uredaja	Broj mape
C2	Type Approval Classification Certificate 03-001315/030990		1
C4	Works Certificate MANUFACT.CERT.	051421249-051421252	1
C6	EX Certificate IMQ 09 ATEX 013 X		1

Naziv knjiga uputa	Map no.
MAGNETIC LEVEL SWITCHES	4

Tehnički podaci

Naziv datoteke (PDM/UCM): 503-8c42-1106522-o-1.pdf

**ULJANIK****STROJARSKA OPREMA - TEHNIČKI
PODACI**Nacrt
1-106-522OJ
8C42Gradnja **503**

Str 27/47, 16.05.2014

Sistem **790882**
PRESOSTATI I TERMOSTATI

Oznaka	Naziv opreme	JM	Kol.
790882-2-100	PROTOCNA SKLOPKA DN100 (4");G=1"	KOM	1

Dobavljač **INEL-MAR D.O.O.**
RUKAVAC 192
51211 MATULJI
HRV Republika Hrvatska
tel. 098/259-998 fax.
Broj i datum narudžbe: **296195** / 30.10.2013

Proizvođač **INEL-MAR D.O.O.**
RUKAVAC 192
51211 MATULJI
HRV Republika Hrvatska
tel. 098/259-998 telefax.

Cert	Tip certifikata Broj certifikata	Clas. Broj uredaja	Broj mape
C2	Type Approval Classification Certificate 89 824 - 94 HH		
C4	Works Certificate EN 10204/2.1		1

Naziv knjiga uputa	Map no.
PADDLE FLOW SWITCH	4

Tehnički podaci

Naziv datoteke (PDM/UCM): 503-8c42-1106522-o-1.pdf

**ULJANIK****STROJARSKA OPREMA - TEHNIČKI
PODACI**Nacrt
1-106-522Gradnja **503**OJ
8C42

Str 28/47, 16.05.2014

Oznaka 790882-2-107	Naziv opreme PRESSURE TRANSMITTER 1-16 BAR	JM KOM	Kol. 2
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Dobavljač **WIKA CROATIA D.O.O.**
Hrastovička 19
10250 ZAGREB-LUČKO
HRV Republika Hrvatska
tel. 01/6531-034 fax.
Broj i datum narudžbe: **298027** / 13.11.2013

Proizvođač **WIKA CROATIA D.O.O.**
Hrastovička 19
10250 ZAGREB-LUČKO
HRV Republika Hrvatska
tel. 01/6531-034 telefax.

Cert	Tip certifikata Broj certifikata	Clas. Broj uredaja	Broj mape
C2	Type Approval Classification Certificate 99179-96HH		1
C4	Works Certificate EN 10.204-2.2		1

Naziv knjiga uputa PRESSURE TRANSMITER	Map no. 4
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Tehnički podaci

Pressure transmitter
-RANGE.... 1-16 bar
-with bracket and testing three way valve
-All process connection(for sensor thread and testing thread)
G 1/2"
-Output 4...20mA(two wire ,external supply)
-Ambient temp + 55 deg C
-Medium temp.. max 80 DEG C
-IP 65
-High vibration environment
-Medium AIR
HRB CERTIFICATE

Naziv datoteke (PDM/UCM): 503-8c42-1106522-o-1.pdf

**ULJANIK****STROJARSKA OPREMA - TEHNIČKI
PODACI**Nacrt
1-106-522OJ
8C42Gradnja **503**

Str 29/47, 16.05.2014

Sistem **801171**
CENTRIFUGALNE PUMPE - BALAST

Oznaka	Naziv opreme	JM	Kol.
801171-2-001	PUMPA KALJUŽE/BALASTA BR.1 70 M3/H; 2 BA	KOM	1

Dobavljač **TTK - TVORNICA TURBINA D.O.O.**
 Vrazova 53, P.P. 168
 47000 KARLOVAC
 HRV Republika Hrvatska
 tel. 047/648-571 fax.
 Broj i datum narudžbe: **293759** / 03.07.2013

Proizvođač **TTK - TVORNICA TURBINA D.O.O.**
 Vrazova 53, P.P. 168
 47000 KARLOVAC
 HRV Republika Hrvatska
 tel. 047/648-571 telefax.

Cert	Tip certifikata Broj certifikata	Clas. Broj uredaja	Broj mape
C1	Classification Certificate 1153-3/296988	HRB 6421	1
C4	Works Certificate 1153-3/296988		1

Naziv knjiga uputa	Map no.
PUMPE STROJARNICE-FINALNA DOKUMENT	1

Tehnički podaci

CENTRIFUGAL PUMP WITH EL. MOTOR

MARK:..... 801171-2-001
 PIECES:..... 1 (one)
 CAPACITY:..... 70 m3/h
 TOTAL MANOMETRIC HEAD:..... 2 Bar
 SERVICE:..... bilge/balast/seawater
 DESCRIPTION: vertical, with intermediate distance shaft
 SPEED..... * rpm
 SELPRIMING..... YES
 LIQUID:..... bilge water, seawater
 SPECIFIC GRAVITY: 1025,4 kg/m3
 KIN. VISCOSITY:..... 1 mm2/s
 WORKING TEMPERATURE: max. 32 deg.C
 NPSH available 4,5 mlc
 MATERIAL OF CASING..... bronze
 MATERIAL OF impeller..... bronze
 MATERIAL OF SHAFT:..... stainless steel
 BEARINGS:..... ball bearings

**ULJANIK****STROJARSKA OPREMA - TEHNIČKI
PODACI**Nacrt
1-106-522OJ
8C42Gradnja **503**

Str 30/47, 16.05.2014

Oznaka	Naziv opreme	JM	Kol.
801171-2-001	PUMPA KALJUŽE/BALASTA BR.1 70 M3/H; 2 BA	KOM	1

SUCTION FLANGE acc.to DIN 2533; DN * / NP 16
DELIVERY FLANGE acc.to DIN 2533; DN * / NP 16

WEIGHT OF PUMP WITH EL. MOTOR:.....*.. kg
CERTIFICATE FOR PUMPCRS
CERTIFICATE FOR EL. MOTOR:..... manufacturer's
INSULATION CLASS :..... IP 55
PROTECTION : F
EL. MOTOR EFFICIENCY CLASS: IE3
ANTICONDENZATING HEATER : NO

**ULJANIK****STROJARSKA OPREMA - TEHNIČKI
PODACI**Nacrt
1-106-522OJ
8C42Gradnja **503**

Str 31/47, 16.05.2014

Oznaka	Naziv opreme	JM	Kol.
801171-2-002	PUMPA KALJUŽE/BALASTA BR.2 70 M3/H; 2 BA	KOM	1

Dobavljač **TTK - TVORNICA TURBINA D.O.O.**

Vrazova 53, P.P. 168

47000 KARLOVAC

HRV Republika Hrvatska

tel. 047/648-571 fax.

Broj i datum narudžbe: **293759** / 03.07.2013Proizvođač **TTK - TVORNICA TURBINA D.O.O.**

Vrazova 53, P.P. 168

47000 KARLOVAC

HRV Republika Hrvatska

tel. 047/648-571 telefax.

Cert	Tip certifikata Broj certifikata	Clas. Broj uredaja	Broj mape
C1	Classification Certificate 1153-3/296988	HRB 6422	1
C4	Works Certificate 1153-3/296988		1

Naziv knjiga uputa	Map no.
PUMPE STROJARNICE-FINALNA DOKUMENT	1

Tehnički podaci

CENTRIFUGAL PUMP WITH EL. MOTOR

MARK:..... 801171-2-002

PIECES:..... 1 (one)

CAPACITY:..... 70 m3/h

TOTAL MANOMETRIC HEAD:..... 2 Bar

SERVICE:..... bilge/balast/seawater

DESCRIPTION: vertical, with intermediate distance shaft

SPEED:..... * rpm

SELPRIMING:..... YES

LIQUID:..... bilge water, seawater

SPECIFIC GRAVITY: 1025,4 kg/m3

KIN. VISCOSITY:..... 1 mm2/s

WORKING TEMPERATURE: max. 32 deg.C

NPSH available 4,5 mlc

MATERIAL OF CASING:..... bronze

MATERIAL OF impeller:..... bronze

MATERIAL OF SHAFT:..... stainless steel

BEARINGS:..... ball bearings

SUCTION FLANGE acc.to DIN 2533; DN * / NP 16

DELIVERY FLANGE acc.to DIN 2533; DN * / NP 16

Naziv datoteke (PDM/UCM): 503-8c42-1106522-o-1.pdf

**ULJANIK****STROJARSKA OPREMA - TEHNIČKI
PODACI**Nacrt
1-106-522OJ
8C42Gradnja **503**

Str 32/47, 16.05.2014

Oznaka	Naziv opreme	JM	Kol.
801171-2-002	PUMPA KALJUŽE/BALASTA BR.2 70 M3/H; 2 BA	KOM	1

WEIGHT OF PUMP WITH EL. MOTOR:.....*.. kg
CERTIFICATE FOR PUMPCRS
CERTIFICATE FOR EL. MOTOR:..... manufacturer's
INSULATION CLASS :..... IP 55
PROTECTION : F
EL. MOTOR EFFICIENCY CLASS: IE3
ANTICONDENZATING HEATER : NO

**ULJANIK****STROJARSKA OPREMA - TEHNIČKI
PODACI**Nacrt
1-106-522Gradnja **503**OJ
8C42

Str 33/47, 16.05.2014

Sistem **801262**
UPRAVLJANJE VENTILIMA,DALJ.HIDR. - SISTEM BALASTA

Oznaka	Naziv opreme	JM	Kol.
801262-2-001	RAVNO ZAPORNI VENT. S ELEKTRO-HIDR.AKT D	PCS	2

Dobavljač **DAMCOS A/S**
Aaderupvej 41
4700 NAESTVED
DNK Danska
tel. +45 55 78 72 00 fax.
Broj i datum narudžbe: **293948** / 06.08.2013

Proizvođač **DAMCOS A/S**
Aaderupvej 41
4700 NAESTVED
DNK Danska
tel. +45 55 78 72 00 telefax.

Cert	Tip certifikata Broj certifikata	Clas. Broj uredaja	Broj mape
C1	Classification Certificate HAM-13-1682-1	HRB	1

Naziv knjiga uputa	Map no.
DAMCOS	5

Tehnički podaci

Adequate electro-hydraulic stand alone unit with manual pump for emergency operation of valve will be delivered. Signalisation open/close with limit switches or similar position feed back unit (mentioned equipment will be installed above freeboard deck).
Valve's local power unit and position indicator enclosure rating IP68.
For NP 16 flanges.
Mark of valve BA-1,BA-2
Control of valves from ICMS at navigation bridge.

**ULJANIK****STROJARSKA OPREMA - TEHNIČKI
PODACI**Nacrt
1-106-522OJ
8C42Gradnja **503**

Str 34/47, 16.05.2014

Sistem **803171**
CENTRIFUGALNE PUMPE - KALJUŽA

Oznaka	Naziv opreme	JM	Kol.
803171-2-012	PUMPA KALJUŽE/PROTUPOŽARA 90M3/H@7BAR;12	KOM	1

Dobavljač **TTK - TVORNICA TURBINA D.O.O.**
 Vrazova 53, P.P. 168
 47000 KARLOVAC
 HRV Republika Hrvatska
 tel. 047/648-571 fax.
 Broj i datum narudžbe: **293759** / 03.07.2013

Proizvođač **TTK - TVORNICA TURBINA D.O.O.**
 Vrazova 53, P.P. 168
 47000 KARLOVAC
 HRV Republika Hrvatska
 tel. 047/648-571 telefax.

Cert	Tip certifikata Broj certifikata	Clas. Broj uredaja	Broj mape
C1	Classification Certificate 1153-3/296988	HRB 6434	1
C4	Works Certificate 1153-3/296988		1

Naziv knjiga uputa	Map no.
PUMPE STROJARNICE-FINALNA DOKUMENTACIJA	1

Tehnički podaci

CENTRIFUGAL PUMP WITH EL. MOTOR

The pump must be with 2 (Two) working points specified below.

MARK:..... 803171-2-012
 PIECES:..... 1 (one)
 CAPACITY,TOTAL MANOMETRIC HEAD:..... 90 m3/h at 7 bar; max. 120 m3/h at 2 bar
 SERVICE:..... bilge/ fire fighting
 DESCRIPTION: vertical, with intermediate distance shaft
 SPEED..... * rpm
 SELPRIMING..... YES
 LIQUID:..... bilge water, seawater
 SPECIFIC GRAVITY: 1025,4 kg/m3
 KIN. VISCOSITY:..... 1 mm2/s
 WORKING TEMPERATURE: max. 32 deg.C
 NPSH available 4,5 mlc
 MATERIAL OF CASING..... bronze
 MATERIAL OF impeller..... bronze
 MATERIAL OF SHAFT:..... stainless steel
 BEARINGS:..... ball bearings

**ULJANIK****STROJARSKA OPREMA - TEHNIČKI
PODACI**Nacrt
1-106-522OJ
8C42Gradnja **503**

Str 35/47, 16.05.2014

Oznaka	Naziv opreme	JM	Kol.
803171-2-012	PUMPA KALJUŽE/PROTUPOŽARA 90M3/H@7BAR;12	KOM	1

SUCTION FLANGE acc.to DIN 2533; DN * / NP 16
DELIVERY FLANGE acc.to DIN 2533; DN * / NP 16

WEIGHT OF PUMP WITH EL. MOTOR:.....*.. kg
CERTIFICATE FOR PUMPCRS
CERTIFICATE FOR EL. MOTOR:..... manufacturer's
INSULATION CLASS :..... IP 55
PROTECTION : F
EL. MOTOR EFFICIENCY CLASS: IE3
ANTICONDENZATING HEATER : NO

**ULJANIK****STROJARSKA OPREMA - TEHNIČKI
PODACI**Nacrt
1-106-522OJ
8C42Gradnja **503**

Str 36/47, 16.05.2014

Sistem **803177**
PUMPE MEMBRANSKE - KALJUŽA

Oznaka	Naziv opreme	JM	Kol.
803177-2-010	MEMBRANSKA KALJUŽNA PUMPA-KRMA 10 M3/H ;	KOM	1

Dobavljač **ABEL GMBH & CO. KG**
 Abel-Twiete 1 D
 21514 BUCHEN
 DEU Germany
 tel. +49 (4155) 818-460 fax.
 Broj i datum narudžbe: **294031** / 08.08.2013

Proizvođač **ABEL GMBH & CO. KG**
 Abel-Twiete 1 D
 21514 BUCHEN
 DEU Germany
 tel. +49 (4155) 818-460 telefax.

Cert	Tip certifikata	Clas.	Broj mape
	Broj certifikata	Broj uredaja	
C1	Classification Certificate QUALITY TEST CERTIFICATE	HRB	1
C4	Works Certificate EC declaration of conformity		

Naziv knjiga uputa	Map no.
EL.DIAPHRAGM PUMPS	5

Tehnički podaci

The bilge pump is electric driven diaphragm pump with a capacity of 10 m3/hr, at a pressure of 2 bar.

Materials:

Pumphousing: cast aluminium

Diaphragm: rubber

Diaphragm plate: stainless steel

Piston: stainless steel.

DIAPHRAGM PUMP WITH ELECTRIC DRIVEN MOTOR - TECHNICAL DATA
 (all data marked with asterisk should be given by the manufacturer)

-
1. MARK:.....803177-2-010
 2. PIECE:.....one (1)
 3. SERVICE:..... bilge
 4. PUMP DESCRIPTION:*
 6. LIQUID:..... sand/sea water. The pump is suitable for particle size of about 15 mm.
 7. SPECIFIC GRAVITY:1025,4 kg/m3
 9. WORKING TEMPERATURE:abt. 45 deg. C.
 - 10.CAPACITY:..... 10 m3/h
 - 11.TOTAL MANOMETRIC HEAD:....2 bar (pressure 1,5bar ; suction -0,5bar min.)

**ULJANIK****STROJARSKA OPREMA - TEHNIČKI
PODACI**Nacrt
1-106-522OJ
8C42Gradnja **503**

Str 37/47, 16.05.2014

Oznaka	Naziv opreme	JM	Kol.
803177-2-010	MEMBRANSKA KALJUŽNA PUMPA-KRMA 10 M3/H ;	KOM	1

12. NPSH available..... M.L.C.
If is liquid level under pump.
13. SUCTION HEAD (PRESSURE):..... - bar
If is liquid level above pump.
Difference between geodetic suction head and friction losses in suction pipe.
14. NPSH:.....*.. MLC
15. TYPE:.....*..
16. AVAILABLE AIR :.....working air ; 7,0 bar
17. NUMBER OF CYCLES PER MINUTE*
18. AIR CONSUMPTION.....*
19. SUCTION FLANGE acc.to DIN 2633;DN * / NP 16
20. DELIVERY FLANGE acc.to DIN 2633;DN * / NP 16
21. GAUGE CONNECTION INCLUDED:.....*..
22. WEIGHT OF PUMP WITH PNEUM. MOTOR:.....*.. kg
23. CERTIFICATE FOR PUMP AND PNEUM. MOTOR ACCORDING TO C.R.S.

**ULJANIK****STROJARSKA OPREMA - TEHNIČKI
PODACI**Nacrt
1-106-522OJ
8C42Gradnja **503**

Str 38/47, 16.05.2014

Oznaka	Naziv opreme	JM	Kol.
803177-2-011	MEMBRANSKA KALJUŽNA PUMPA-PRAMAC 10 M3/H	KOM	1

Dobavljač **ABEL GMBH & CO. KG**
 Abel-Twiete 1 D
 21514 BUCHEN
 DEU Germany
 tel. +49 (4155) 818-460 fax.
 Broj i datum narudžbe: **294031** / 08.08.2013

Proizvođač **ABEL GMBH & CO. KG**
 Abel-Twiete 1 D
 21514 BUCHEN
 DEU Germany
 tel. +49 (4155) 818-460 telefax.

Cert	Tip certifikata Broj certifikata	Clas. Broj uredaja	Broj mape
C1	Classification Certificate QUALITY TEST CERTIFICATE	HRB	1
C4	Works Certificate		

Naziv knjiga uputa	Map no.
EL.DIAPHRAGM PUMP	5

Tehnički podaci

The bilge pump is electric driven diaphragm pump with a capacity of 10 m3/hr. at a pressure of 2 bar.
 Materials:
 Pumphousing: cast aluminium
 Diaphragm: rubber
 Diaphragm plate: stainless steel
 Piston: stainless steel.

DIAPHRAGM PUMP WITH ELECTRIC DRIVEN MOTOR - TECHNICAL DATA
 (all data marked with asterisk should be given by the manufacturer)

-
- MARK:.....803177-2-010
 - PIECE:.....one (1)
 - SERVICE:..... bilge
 - PUMP DESCRIPTION:*
 - LIQUID:..... sand/sea water. The pump is suitable for particle size of about 15 mm.
 - SPECIFIC GRAVITY:1025,4 kg/m3
 - WORKING TEMPERATURE:abt. 45 deg. C.
 - CAPACITY:..... 10 m3/h
 - TOTAL MANOMETRIC HEAD:....2 bar (pressure 1,5bar ; suction -0,5bar min.)
 - NPSH available..... M.L.C.
If is liquid level under pump.
 - SUCTION HEAD (PRESSURE):..... - bar

**ULJANIK****STROJARSKA OPREMA - TEHNIČKI
PODACI**Nacrt
1-106-522OJ
8C42Gradnja **503**

Str 39/47, 16.05.2014

Oznaka	Naziv opreme	JM	Kol.
803177-2-011	MEMBRANSKA KALJUŽNA PUMPA-PRAMAC 10 M3/H	KOM	1

If is liquid level above pump.

Difference between geodetic suction head and friction losses in suction pipe.

14. NPSH:.....*.. MLC

15. TYPE:.....*..

16. AVAILABLE AIR :.....working air ; 7,0 bar

17. NUMBER OF CYCLES PER MINUTE*

18. AIR CONSUMPTION.....*

19. SUCTION FLANGE acc.to DIN 2633;DN * / NP 16

20. DELIVERY FLANGE acc.to DIN 2633;DN * / NP 16

21. GAUGE CONNECTION INCLUDED:.....*..

22. WEIGHT OF PUMP WITH PNEUM. MOTOR:.....*.. kg

23. CERTIFICATE FOR PUMP AND PNEUM. MOTOR ACCORDING TO C.R.S.

**ULJANIK****STROJARSKA OPREMA - TEHNIČKI
PODACI**Nacrt
1-106-522OJ
8C42Gradnja **503**

Str 40/47, 16.05.2014

Sistem **813171**
CENTRIFUGALNE PUMPE - PROTUPOŽAR, PRANJE MV

Oznaka	Naziv opreme	JM	Kol.
813171-2-101	PUMPA PROTUPOŽARA BR.2 90 M3/H; 7 BAR	KOM	1

Dobavljač **TTK - TVORNICA TURBINA D.O.O.**
 Vrazova 53, P.P. 168
 47000 KARLOVAC
 HRV Republika Hrvatska
 tel. 047/648-571 fax.
 Broj i datum narudžbe: **293759** / 03.07.2013

Proizvođač **TTK - TVORNICA TURBINA D.O.O.**
 Vrazova 53, P.P. 168
 47000 KARLOVAC
 HRV Republika Hrvatska
 tel. 047/648-571 telefax.

Cert	Tip certifikata Broj certifikata	Clas. Broj uredaja	Broj mape
C1	Classification Certificate 1153-3/296988	HRB 6424	1
C4	Works Certificate 1153-3/296988		1

Naziv knjiga uputa	Map no.
PUMPE STROJARNICE-FINALNA DOKUMENTACIJA	1

Tehnički podaci

CENTRIFUGAL PUMP WITH EL. MOTOR

MARK:..... 813171-2-101
 PIECES:..... 1 (one)
 CAPACITY:..... 90 m3/h
 TOTAL MANOMETRIC HEAD:..... 7 Bar
 SERVICE:..... sea water, fire fighting, drencher
 DESCRIPTION: vertical, with intermediate distance shaft
 SPEED..... * rpm
 SELPRIMING..... YES
 LIQUID:..... sea water
 SPECIFIC GRAVITY: 1025,4 kg/m3
 KIN. VISCOSITY:..... 1 mm2/s
 WORKING TEMPERATURE: max. 32 deg.C
 NPSH available 4,5 mlc
 MATERIAL OF CASING..... bronze
 MATERIAL OF impeller..... bronze
 MATERIAL OF SHAFT:..... stainless steel
 BEARINGS:..... ball bearings

**ULJANIK****STROJARSKA OPREMA - TEHNIČKI
PODACI**Nacrt
1-106-522OJ
8C42Gradnja **503**

Str 41/47, 16.05.2014

Oznaka	Naziv opreme	JM	Kol.
813171-2-101	PUMPA PROTUPOŽARA BR.2 90 M3/H; 7 BAR	KOM	1

SUCTION FLANGE acc.to DIN 2533; DN * / NP 16
DELIVERY FLANGE acc.to DIN 2533; DN * / NP 16

WEIGHT OF PUMP WITH EL. MOTOR:.....*.. kg
CERTIFICATE FOR PUMPCRS
CERTIFICATE FOR EL. MOTOR:..... manufacturer's
INSULATION CLASS :..... IP 55
PROTECTION : F
EL. MOTOR EFFICIENCY CLASS: IE3
ANTICONDENZATING HEATER : NO

**ULJANIK****STROJARSKA OPREMA - TEHNIČKI
PODACI**Nacrt
1-106-522OJ
8C42Gradnja **503**

Str 42/47, 16.05.2014

Oznaka	Naziv opreme	JM	Kol.
813171-2-103	PUMPA PROTUPOŽARA BR.3 90 M3/H; 7 BAR	KOM	1

Dobavljač **TTK - TVORNICA TURBINA D.O.O.**

Vrazova 53, P.P. 168

47000 KARLOVAC

HRV Republika Hrvatska

tel. 047/648-571 fax.

Broj i datum narudžbe: **293759** / 03.07.2013Proizvođač **TTK - TVORNICA TURBINA D.O.O.**

Vrazova 53, P.P. 168

47000 KARLOVAC

HRV Republika Hrvatska

tel. 047/648-571 telefax.

Cert	Tip certifikata Broj certifikata	Clas. Broj uredaja	Broj mape
C1	Classification Certificate 1153-3/296988	HRB	1
C4	Works Certificate 1153-3/296988		1

Naziv knjiga uputa	Map no.
PUMPE STROJARNICE-FINALNA DOKUMENTACIJA	1

Tehnički podaci

CENTRIFUGAL PUMP WITH EL. MOTOR

MARK:..... 813171-2-103

PIECES:..... 1 (one)

CAPACITY:..... 90 m3/h

TOTAL MANOMETRIC HEAD:..... 7 Bar

SERVICE:..... sea water, fire fighting, drencher

DESCRIPTION: vertical, with intermediate distance shaft

SPEED:..... * rpm

SELPRIMING:..... YES

LIQUID:..... sea water

SPECIFIC GRAVITY: 1025,4 kg/m3

KIN. VISCOSITY:..... 1 mm2/s

WORKING TEMPERATURE: max. 32 deg.C

NPSH available 4,5 mlc

MATERIAL OF CASING:..... bronze

MATERIAL OF impeller:..... bronze

MATERIAL OF SHAFT:..... stainless steel

BEARINGS:..... ball bearings

SUCTION FLANGE acc.to DIN 2533; DN * / NP 16

DELIVERY FLANGE acc.to DIN 2533; DN * / NP 16

Naziv datoteke (PDM/UCM): 503-8c42-1106522-o-1.pdf

**ULJANIK****STROJARSKA OPREMA - TEHNIČKI
PODACI**Nacrt
1-106-522OJ
8C42Gradnja **503**

Str 43/47, 16.05.2014

Oznaka	Naziv opreme	JM	Kol.
813171-2-103	PUMPA PROTUPOŽARA BR.3 90 M3/H; 7 BAR	KOM	1

WEIGHT OF PUMP WITH EL. MOTOR:.....*.. kg
CERTIFICATE FOR PUMPCRS
CERTIFICATE FOR EL. MOTOR:..... manufacturer's
INSULATION CLASS :..... IP 55
PROTECTION : F
EL. MOTOR EFFICIENCY CLASS: IE3
ANTICONDENZATING HEATER : NO

**ULJANIK****STROJARSKA OPREMA - TEHNIČKI
PODACI**Nacrt
1-106-522OJ
8C42Gradnja **503**

Str 44/47, 16.05.2014

Oznaka	Naziv opreme	JM	Kol.
813171-2-104	PUMPA RASPRŠIVANJA 110 M3/H : 7 BAR	KOM	1

Dobavljač **TTK - TVORNICA TURBINA D.O.O.**

Vrazova 53, P.P. 168

47000 KARLOVAC

HRV Republika Hrvatska

tel. 047/648-571 fax.

Broj i datum narudžbe: **296988** / 16.10.2013Proizvođač **TTK - TVORNICA TURBINA D.O.O.**

Vrazova 53, P.P. 168

47000 KARLOVAC

HRV Republika Hrvatska

tel. 047/648-571 telefax.

Cert	Tip certifikata Broj certifikata	Clas. Broj uredaja	Broj mape
C1	Classification Certificate 1153-3/296988	HRB	1
C4	Works Certificate 1153-3/296988		1

Naziv knjiga uputa	Map no.
PUMPE STROJARNICE-FINALNA DOKUMENTACIJA	1

Tehnički podaci

CENTRIFUGAL PUMP WITH EL. MOTOR

MARK:..... 813171-2-104

PIECES:..... 1 (one)

CAPACITY:..... 110 m3/h

TOTAL MANOMETRIC HEAD:..... 7 Bar

SERVICE:..... sea water, fire fighting, drencher

DESCRIPTION: vertical, with intermediate distance shaft

SPEED:..... * rpm

SELPRIMING:..... YES

LIQUID:..... sea water

SPECIFIC GRAVITY: 1025,4 kg/m3

KIN. VISCOSITY:..... 1 mm2/s

WORKING TEMPERATURE: max. 32 deg.C

NPSH available 4,5 mlc

MATERIAL OF CASING:..... bronze

MATERIAL OF impeller:..... bronze

MATERIAL OF SHAFT:..... stainless steel

BEARINGS:..... ball bearings

SUCTION FLANGE acc.to DIN 2533; DN * / NP 16

DELIVERY FLANGE acc.to DIN 2533; DN * / NP 16

**ULJANIK****STROJARSKA OPREMA - TEHNIČKI
PODACI**Nacrt
1-106-522OJ
8C42Gradnja **503**

Str 45/47, 16.05.2014

Oznaka	Naziv opreme	JM	Kol.
813171-2-104	PUMPA RASPRŠIVANJA 110 M3/H : 7 BAR	KOM	1

WEIGHT OF PUMP WITH EL. MOTOR:.....*.. kg
CERTIFICATE FOR PUMPCRS
CERTIFICATE FOR EL. MOTOR:..... manufacturer's
INSULATION CLASS :..... IP 55
PROTECTION : F
EL. MOTOR EFFICIENCY CLASS: IE3
ANTICONDENZATING HEATER : NO

**SADRŽAJ**

Broj sistema. / Oznaka	Naziv opreme	Str.
441143-4-001	BRUSILICA S UGRAĐENIM PRIBOROM 2X 150X20	1
441143	BRUSILICA	1
581171	CENTRIFUGALNE PUMPE - SANITARNA VODA	2
581171-2-001	PUMPA SANITARNE VODE 10 M3/H; 6 BAR	2
631661	SKLOP PROPULZIONE JEDINICE	4
631661-1-010	OKRETNi PROPULZOR, LIJEVOKRETNi	4
631661-1-020	OKRETNi PROPULZOR, DESNOKRETNi	5
701173	PUMPE VIJČANE - PREBACIVANJE GORIVA	6
701173-2-001	PUMPA TRANSFERA GORIVA - PRAMČANA 4 M3/H	6
701173-2-002	PUMPA TRANSFERA GORIVA - KRMENA 4 M3/H;	8
711173	PUMPE VIJČANE - PREBACIVANJE MAZIVA	10
711173-2-001	TRANSFER PUMPA PRLJAVOG ULJA-PRAMČANA 4	10
711173-2-002	TRANSFER PUMPA PRLJAVOG ULJA-KRMENA 4 M3	12
721263	UREĐAJ PROTIV OBRAŠTANJA - AKTIVNI HIPOKLORIT	14
721263-0-001	ELEKTRO PANEL R200	14
733165	KOMPRES.RADNOG ZRAKA,ROTACIONI	15
733165-2-001	KOMPRESOR RADNOG ZRAKA 50M3/H FA.D. / 7	15
790876	TERMOMETRI	17
790876-2-023	TERMOMETAR RAVNI 0-60°C,G=1/2,L1=63MM	17
790876-2-130	TERMOMETAR CAPILARNI 0-60 C,G=1/2,L1=160	18
790881	NIVOSTATI	19
790881-2-120	NIVOSTAT	19
790881-2-121	NIVOSTAT	20
790881-2-128	NIVOSTAT	21
790881-2-129	NIVOSTAT	22
790881-2-160	NIVOSTAT DVOKONTAKTNI	23
790881-2-161	NIVOSTAT DVOKONTAKTNI-DUZI	24
790881-2-190	NIVOSTAT WITH PROTECTION TUBE	25
790881-2-191	ZENER BARIJERA	26
790882	PRESOSTATI I TERMOSTATI	27
790882-2-100	PROTOKNA SKLOPKA DN100 (4");G=1"	27
790882-2-107	PRESSURE TRANSMITTER 1-16 BAR	28
801171	CENTRIFUGALNE PUMPE - BALAST	29
801171-2-001	PUMPA KALJUŽE/BALASTA BR.1 70 M3/H; 2 BA	29

**SADRŽAJ**

Broj sistema. / Oznaka	Naziv opreme	Str.
801171-2-002	PUMPA KALJUŽE/BALASTA BR.2 70 M3/H; 2 BA	31
801262	UPRAVLJANJE VENTILIMA,DALJ.HIDR. - SISTEM BALASTA	33
801262-2-001	RAVNO ZAPORNI VENT. S ELEKTRO-HIDR.AKT D	33
803171	CENTRIFUGALNE PUMPE - KALJUŽA	34
803171-2-012	PUMPA KALJUŽE/PROTUPOŽARA 90M3/H@7BAR;12	34
803177	PUMPE MEMBRANSKE - KALJUŽA	36
803177-2-010	MEMBRANSKA KALJUŽNA PUMPA-KRMA 10 M3/H ;	36
803177-2-011	MEMBRANSKA KALJUŽNA PUMPA-PRAMAC 10 M3/H	38
813171	CENTRIFUGALNE PUMPE - PROTUPOŽAR, PRANJE MV	40
813171-2-101	PUMPA PROTUPOŽARA BR.2 90 M3/H; 7 BAR	40
813171-2-103	PUMPA PROTUPOŽARA BR.3 90 M3/H; 7 BAR	42
813171-2-104	PUMPA RASPRŠIVANJA 110 M3/H : 7 BAR	44